

The periphery is enough: a region-ablation audit of the MedMNIST v2 3D CT benchmarks

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Abstract

Three-dimensional CT benchmarks are used to rank volumetric classifiers, but a fixed-size crop centred on a target structure also captures the anatomy around it. We ask how much of the label survives deletion of the target. Working with 1496 crops of 64 by 64 by 64 voxels drawn from the three CT subsets of MedMNIST v2, we split each crop into a central core, the middle 32 voxels of every axis and 12.5 per cent of the volume, and the surrounding periphery, and we fit a regularised linear classifier on mean-pooled intensities from one region at a time, training on the official splits and reporting held-out test performance. The three subsets behave in three different ways. For OrganMNIST3D the periphery alone reaches 73.7 per cent accuracy and the outermost rim alone reaches 75.4 per cent (95 per cent confidence interval 68.9 to 82.0), matching the full crop at 74.3 per cent and far exceeding the core at 55.1 per cent. For NoduleMNIST3D the core dominates, with an area under the curve of 0.834 against 0.849 for the full crop, but the periphery still ranks above chance at 0.679 (0.561 to 0.789), and a single scalar, the high-attenuation volume fraction of the core, recovers 0.758, which is 89.3 per cent of the full-crop value. For FractureMNIST3D our decoder has no power at all, 34.1 per cent with an interval of 24.7 to 43.5 that spans chance, and the data explain why: only 4.7 per cent of peripheral voxels are non-zero in these bone-masked crops. Accuracy on these benchmarks is therefore not by itself evidence that a model reads the structure the benchmark is named after, and a periphery-only baseline is cheap enough to report alongside it.

1 Introduction

Deep networks now read three-dimensional CT well enough to be evaluated against radiologists in lung cancer screening [Ardila et al., 2019], and the benchmarks that rank such models are increasingly of the lightweight, standardised kind. MedMNIST v2 is the most widely used of these: it packages several clinical CT sources into fixed-size voxel crops with fixed splits, so a new architecture can be compared without touching the original scans [Yang et al., 2023]. Three of its subsets are CT volumes: lung nodule crops derived from LIDC-IDRI [Armato et al., 2011], organ crops derived from the Liver Tumor Segmentation benchmark [Bilic et al., 2022], and rib segment crops derived from a rib fracture cohort [Jin et al., 2020].

A fixed-size crop centred on a target structure is a convenient abstraction with a known hazard: it also contains everything around the target. Networks are opportunistic and will use whatever predicts the label [Geirhos et al., 2020]. In radiology this has repeatedly meant reading the hospital, the scanner or the patient rather than the disease [Zech et al., 2018, Gichoya et al., 2022], and such failures are now treated as a systematic methodological problem for medical imaging benchmarks rather than as isolated accidents [Varoquaux and Cheplygina, 2022, Kapoor and Narayanan, 2023]. What has been examined less often for the small volumetric benchmarks is the simplest form of the question: where inside the field of view does the label information actually sit?

We arrived at this question by looking at the data. We rendered 8 of the volumes as three orthogonal central slices and three maximum-intensity projections and read the renders before choosing any method. In the nodule crops we could frequently not point to a lesion at all: the field was chest wall, pleura and the vessel tree, and the two crops in which a mass was obvious were both labelled

malignant and both large. In the organ crops the nominal organ shared the field with neighbouring viscera, the body outline and the spine. In the rib crops two or three intact ribs shared the field with the fractured one. In every case the target structure was a minority of what we could see.

We therefore ran a region ablation. We delete the central core of each crop, keep only the periphery, and ask how much of the label a simple decoder can still recover; we then invert the mask and ask the same of the core. The decoder is deliberately weak, a linear model on mean-pooled intensities, so that whatever it recovers is a lower bound on what is available. Our contributions are: a region-ablation audit of all three CT subsets under their official splits, with permutation, intensity-normalisation and bootstrap controls; the finding that organ identity is fully recoverable from the periphery alone while nodule malignancy is mostly, though not entirely, carried by the core; the finding that a single size-proxy scalar accounts for most of the nodule signal; and an honest null for the fracture subset together with the measurement that explains it.

2 Data

We work with 1496 CT volume crops, each 64 by 64 by 64 voxels stored as 8-bit intensities spanning 0 to 255, drawn from the three CT subsets of MedMNIST v2 [Yang et al., 2023]. There are 500 nodule crops labelled benign or malignant, balanced at 250 each; 495 organ crops over 11 organ classes with 45 crops per class; and 501 rib segment crops over 3 fracture categories with 167 each. We keep the official split assignment that ships with the data: 362 training, 50 validation and 88 test volumes for the nodule subset, 269, 59 and 167 for the organ subset, and 377, 39 and 85 for the fracture subset. No volume moves between splits and the test split is touched once, at the end.

Before any modelling we inspected 8 volumes visually, chosen to span the subsets and the classes, using rendered projections rather than summary statistics. Each render shows three orthogonal central slices and three maximum-intensity projections. The slice panels supported perception in every case we looked at: dark lung parenchyma speckled with vessels seen end on, a lobulated bright mass with a sharp margin against lung, a homogeneous liver with a curved capsule beside gas-filled bowel, a femoral head with a bright cortical rim inside grey muscle, and rib cross sections with a bright cortex around a darker medulla. The projection panels were less reliable and this is worth stating plainly. For the bone-masked rib crops they were the most informative view, showing two or three rib bands crossing the field with a local knot of dense texture or an abrupt step where the fracture is. For crops containing bone or iodinated contrast they saturated: the femur projections were almost uniformly white and the liver projections were white with dark stripes, with no readable organ boundary in either. We based our reading on the slice panels and treat maximum-intensity projections of high-attenuation anatomy as unreliable.

The visual impression that drove the rest of this paper is that these crops are context-rich. The nodule crops routinely contain chest wall and vessels; the organ crops contain the body outline and the neighbouring viscera; the rib crops contain intact neighbouring ribs. Whether that context is merely present or is actually sufficient is a quantitative question, which is what we test next.

3 Analysis

Every volume is reduced to an 8 by 8 by 8 grid of mean-pooled cells, each cell averaging an 8 by 8 by 8 block of voxels, giving 512 cells per volume. Regions are defined on this grid and on the voxel grid consistently. CORE is the central 32 voxels of each axis, that is 64 cells and 12.5 per cent of the volume. SHELL is its complement, 448 cells and 87.5 per cent of the volume. RIM is the outermost shell of thickness 8 voxels, everything outside the inner box of side 48, which is 296 cells and 57.8 per cent of the volume; it shares no voxel with the core and no boundary with it either. GLOBAL is a control that discards spatial layout entirely and keeps only four scalars of the whole volume.

The feature vector for a region is the pooled values of its cells together with four intensity statistics computed over that region at full resolution: mean, standard deviation, the ninetieth percentile, and the fraction of voxels above an attenuation threshold of 150. This gives 516 features for FULL, 68 for CORE, 452 for SHELL, 300 for RIM and 4 for GLOBAL. Features are standardised on the

training data. The classifier is multinomial logistic regression with an L2 penalty whose strength is chosen on the official validation split from a grid of 4 values between 0.01 and 10.0; the model is then refit on training plus validation and evaluated once on the test split. We report accuracy for all subsets and, for the binary nodule task, the area under the receiver operating characteristic curve.

Three controls guard the interpretation. First, a label permutation: the same pipeline is refit on the periphery with shuffled training labels, which should collapse to chance and does. Second, per-volume standardisation: every volume is z-scored before pooling, which removes any per-scan offset in overall brightness or dose, so that a surviving result cannot be attributed to global intensity. Third, occupancy: we measure what fraction of voxels are non-zero in each region of each subset, so that a null result can be attributed to absent signal rather than left unexplained. Confidence intervals come from 2000 bootstrap resamples of the test split.

4 Results

The three subsets behave in three different ways, and only one of them behaves the way the name of the benchmark implies. Organ identity is recoverable from the periphery alone at the accuracy of the full crop. Nodule malignancy is mostly carried by the core, but the periphery still ranks above chance. Fracture category is not recoverable by our decoder from any region.

Table 1. Test-split performance by region. Accuracies are percentages, with bootstrap 95% confidence intervals in brackets where they were computed; the AUC column is the binary benign-versus-malignant task. Uniform chance accuracy is 50.0, 9.1 and 33.3 for the three subsets.

Region	Volume used (%)	Nodule acc.	Nodule AUC	Organ acc.	Fracture acc.
FULL	all	76.1 (67.0–85.2)	0.849 (0.762–0.921)	74.3 (67.7–80.8)	34.1 (24.7–43.5)
CORE	12.5	75.0 (65.9–84.1)	0.834 (0.741–0.913)	55.1 (47.9–62.9)	40.0 (30.6–50.6)
SHELL	87.5	62.5 (52.3–72.7)	0.679 (0.561–0.789)	73.7 (66.5–80.2)	24.7 (16.5–34.1)
RIM	57.8	59.1 (48.9–69.3)	0.587 (0.465–0.704)	75.4 (68.9–82.0)	32.9 (23.5–43.5)
GLOBAL	all	56.8	0.633	46.1	31.8
SHELL, permuted labels	87.5	54.5		8.4	25.9
Majority class		55.7		12.0	37.6

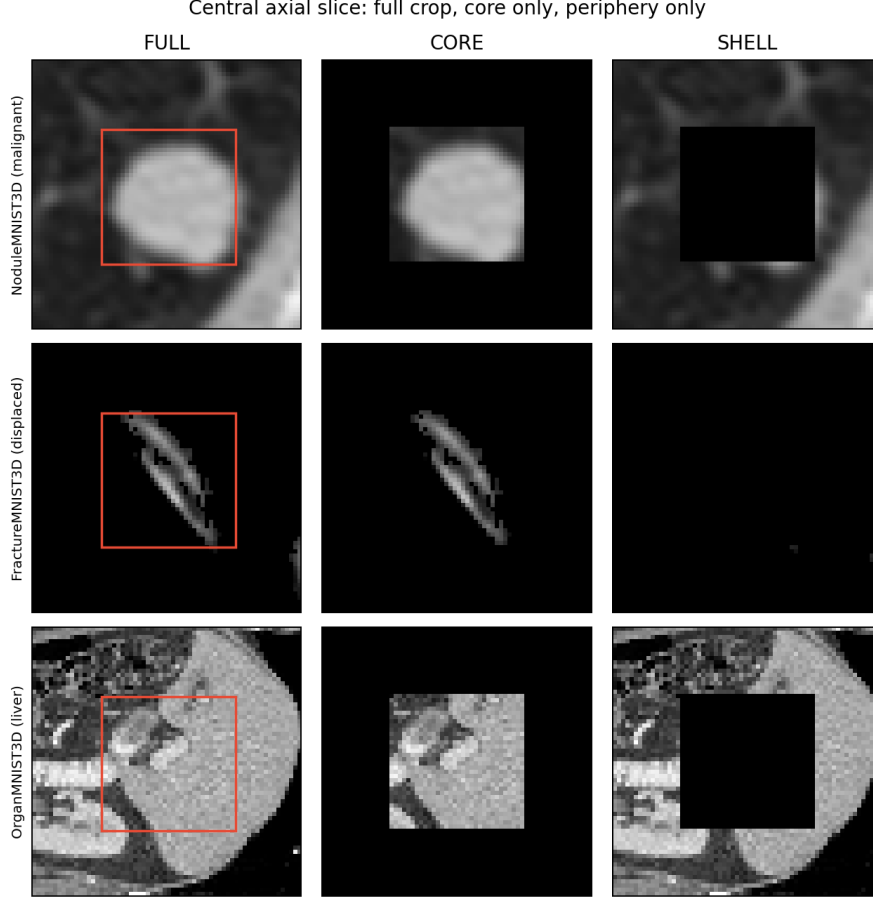


Figure 1. Central axial slice of one crop from each subset, shown in full, restricted to the central core, and restricted to the periphery. The red box marks the core. The nodule sits inside the core and is removed cleanly by the mask, while the periphery keeps the chest wall and the vessel tree. The liver crop keeps its body outline, subcutaneous fat, bowel gas and spine in the periphery. The bone-masked rib crop keeps almost nothing at all, which is why the fracture decoder has nothing to work with outside the core.

In OrganMNIST3D the periphery is not a nuisance, it is the signal. Deleting the central core and keeping the outer 87.5 per cent of the volume gives 73.7 per cent test accuracy (66.5 to 80.2), statistically indistinguishable from the 74.3 per cent (67.7 to 80.8) of the full crop, against a uniform chance level of 9.1 per cent. Pushing further, the outermost rim alone, which touches no voxel within 8 of the core and covers 57.8 per cent of the volume, gives 75.4 per cent (68.9 to 82.0). The core alone, which is where the nominal organ centre is, gives 55.1 per cent (47.9 to 62.9), and its interval does not overlap the rim interval. The result is not an artefact of the fitting procedure: with permuted training labels the same periphery features give 8.4 per cent, at chance. Nor is it an artefact of overall brightness: after per-volume standardisation the rim still gives 74.9 per cent and the periphery 76.0 per cent, while the four global intensity scalars on their own give only 46.1 per cent. Figure 1 shows the mechanism directly. The periphery panel of the liver crop still contains the body outline, the subcutaneous fat, gas-filled bowel and the spine, and knowing which part of the trunk you are looking at narrows 11 organ classes a very long way without ever looking at the organ.

In NoduleMNIST3D the ordering reverses, as Figure 2 shows. The core alone reaches an area under the curve of 0.834 (0.741 to 0.913), essentially the full crop value of 0.849 (0.762 to 0.921), while the periphery alone reaches 0.679 (0.561 to 0.789). That interval excludes the chance value of 0.500, so the periphery of a lung nodule crop does carry malignancy information, but it is a weak leak rather than a sufficient statistic: periphery accuracy of 62.5 per cent sits only just above the majority-

class rate of 55.7 per cent. The outermost rim is weaker still, 0.587 (0.465 to 0.704), an interval that does not exclude chance. Per-volume standardisation does not remove the peripheral signal; it raises it, to 0.722 for the periphery and 0.753 for the rim, which argues that what the periphery carries is anatomy rather than scan brightness. We note this in passing because the same normalisation lowers the full-crop value to 0.742, and we would not build a claim on a single reordering of overlapping intervals.

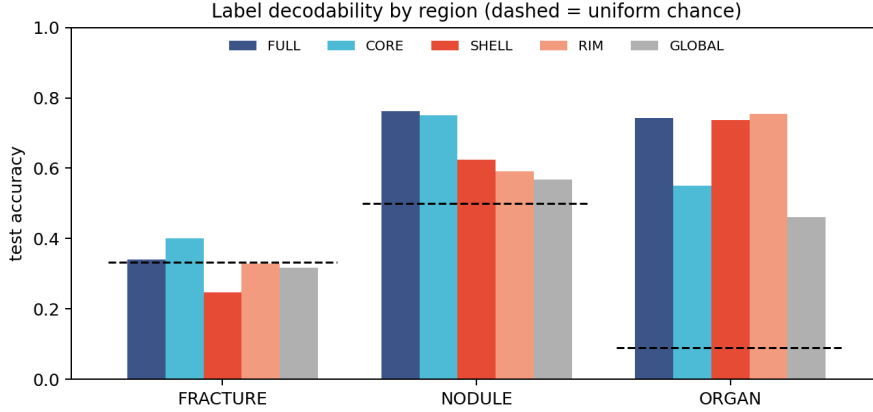


Figure 2. Test-split accuracy by region for the three subsets. Dashed lines are uniform chance. For OrganMNIST3D the periphery and even the outermost rim match or exceed the full crop while the core alone falls far behind; for NoduleMNIST3D the ordering is the opposite; for FractureMNIST3D no region is distinguishable from chance.

What the nodule core carries is close to a single number. The fraction of core voxels above the high-attenuation threshold, a crude proxy for how much solid tissue sits in the middle of the crop, separates the test classes on its own with an area under the curve of 0.758 (Mann-Whitney $p = 3.46 \times 10^{-5}$), with a mean of 0.1126 in benign crops against 0.3008 in malignant ones. That single scalar is 89.3 per cent of the full-crop discriminative performance, and the right panel of Figure 3 shows how much of it comes from a large malignant tail rather than from a clean separation. The same scalar measured on the periphery gives 0.519, that is nothing. This is consistent with clinical experience, where lesion size is the dominant term in nodule malignancy risk [McWilliams et al., 2013], and it means that a model reported on this subset can look strong while doing little more than measuring the middle of the box.

For FractureMNIST3D we report a null. The full crop gives 34.1 per cent (24.7 to 43.5) on a 3-class problem whose chance level is 33.3 per cent and whose majority class rate is 37.6 per cent; no region beats either baseline, and the core, at 40.0 per cent (30.6 to 50.6), is the only one that even nominally exceeds them. Our decoder simply cannot do this task, so we cannot use it to test whether the fracture label leaks into the periphery. The occupancy control says why the periphery half of that question is close to vacuous anyway: these crops are bone-masked, and only 4.7 per cent of peripheral voxels and 4.2 per cent of rim voxels are non-zero, with a mean peripheral intensity of 5.1 out of 255, against 10.8 per cent occupancy in the core. The nodule and organ crops, by contrast, are 99.4 per cent and 89.8 per cent occupied outside the core. The left column of Figure 1 makes the same point visually: masking the core of a rib crop leaves an almost black volume. Grading a rib fracture as buckle, nondisplaced or displaced is a fine geometric judgement about cortical continuity, and mean-pooled intensities over blocks of 8 voxels destroy exactly that.

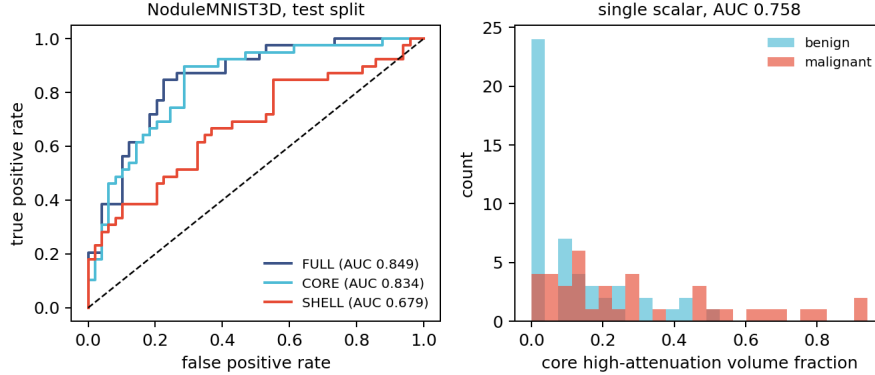


Figure 3. NoduleMNIST3D. Left: receiver operating characteristic curves on the test split for the full crop, the core alone and the periphery alone. Right: distribution of the single scalar we use as a size proxy, the fraction of core voxels above the high-attenuation threshold, for benign and malignant crops in the test split.

5 Discussion

The three subsets fail the same audit in three different ways, and the differences matter more than the average. OrganMNIST3D is, under our decoder, a body-region localisation task wearing the clothes of an organ recognition task: the outer rim of the crop, which contains no part of the organ centre, supports as much accuracy as the whole crop. That does not make the benchmark useless, but it does mean a headline accuracy on it is not evidence that a model has learned organ appearance, and any claim of that kind needs a periphery-only baseline printed next to it. NoduleMNIST3D is better behaved: the core carries most of the signal, as it should. But the periphery is not silent, and the core signal is itself concentrated in something as simple as the amount of solid tissue in the middle of the box, which is a reminder that a strong number on a small benchmark can be produced by a mechanism far shallower than the clinical task it stands for. FractureMNIST3D we cannot judge; we can only report that our decoder has no power on it and that its crops carry almost no peripheral content to leak.

The practical recommendation is cheap. A region ablation of the kind reported here costs a few minutes of CPU time on a benchmark of this size and gives a calibration that accuracy alone cannot: if a periphery-only linear model matches your network, your network’s number is about context. This complements the acquisition-level confounds that dominate the shortcut literature in radiology [Zech et al., 2018, Gichoya et al., 2022, Geirhos et al., 2020], which ask which scans a model is exploiting; the question here is spatial and within-image, and it is answerable without any access to metadata.

Several limitations bound what we may conclude. Our decoder is a linear model on mean-pooled intensities, so every number here is a lower bound on decodability: a convolutional network would very likely extract more from both the core and the periphery, and the fracture null is a statement about our features, not about the subset. The test splits are small, at 88, 167 and 85 volumes, and the confidence intervals are correspondingly wide; the comparison we lean on hardest, rim against core in the organ subset, is the one whose intervals are separated, and we do not lean on the narrower reorderings. We ran one protocol with one random seed, and we did not repeat it over reshuffled splits, so the leave-one-benchmark-out behaviour of these estimates is unknown. We used a subsample of 1496 volumes rather than the complete subsets, so the absolute accuracies are not comparable with published leaderboard values, and the whole audit should be read as being about the structure of the data rather than about anyone’s reported score. The core boundary at 32 voxels is a design choice, and a lesion larger than the core would spill into the periphery and inflate the peripheral result, which is one reason we also report the rim, whose gap from the core is 8 voxels on every side. Finally, the visual reading that motivated the hypothesis was done by one reader on 8 volumes, and the maximum-intensity projections were unusable for bone and contrast-enhanced

anatomy, so the perceptual claims in the Data section rest on the slice panels alone.

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